Sarthak Gupta

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Education

University of Florida — Gainesville, FL
B.S./M.S. in Computer Science, Minor in Economics ***(GPA: 3.67)*** — *Aug. 2024 – Dec. 2028*
Coursework: Data Structures & Algorithms, Operating Systems, Computer Organization, Database Systems, Linear Algebra, Advanced Data Structures, Algorithm Abstraction and Design

Experience

AI Software Engineering Intern — May 2026 – Aug. 2026
Qualcomm — *San Diego, CA*
- Developed an **on-device AI** pipeline for **voice** and **image** processing that runs entirely on **Snapdragon** chips.
- Enabled use cases such as extracting meeting details (date, time, location) from a voice message in any messaging app and adding the event to the user's calendar automatically, without data leaving the device.

Co-Founder & Lead Engineer — Jan. 2026 – Present
Lingua (linguaclassroom.com) | AI-powered LMS for K-12 world language teachers — *Gainesville, FL*
- Shipped a full-stack platform (**Next.js**, **Node.js**, **PostgreSQL**) live in Florida schools with **400+ users** and **3,000+ submissions**; led **FERPA/COPPA** compliance and cleared a school's full technology and data-privacy review.
- Built **Deo**, a teacher-facing AI agent running a bounded **tool-calling loop** over **20+ tools**, with confirmation before any significant action and all-or-nothing creates, cutting setup from about 25 min to **under 90 s**.
- Validated AI output in code before it reaches students: server checks enforce structure and vocabulary coverage, a legibility check gates handwritten work, and scores are computed from per-turn model labels.
- Engineered the **Canvas LMS** integration on **LTI 1.3** (OIDC login, JWT verification, nonce replay protection, AGS grade passback) and moved tenant isolation into **Postgres Row-Level Security** with **40+ policies** over a 31-table schema.

AI Software Developer — Oct. 2025 – May 2026
Florida Community Innovation Foundation @ University of Florida — *Gainesville, FL*
- Designed and deployed **LangChain**-based **agentic pipelines** (**OpenAI APIs**, Python) that autonomously extract, validate, and structure nonprofit and government service data from unstructured web sources.
- Built scraping and data-cleaning pipelines for **5,000+ resource listings**, improving data coverage by **35%**.

Research

Undergraduate Student Researcher — Aug. 2026 – Present
EdgeAI Lab @ University of Florida — *Gainesville, FL*
- Researching a **hardware-in-the-loop (HIL)** extension of NVIDIA's **AI Telco Engineer**, an LLM-agent framework that designs wireless communication protocols in **Sionna** simulation, so candidate designs are evaluated on real radio hardware.
- Validating that the HIL testbed performs as expected by benchmarking hardware results against simulated-channel baselines.

Undergraduate Student Researcher — May 2025 – Present
Virtual Experiences Research Group (VERG) Lab @ University of Florida — *Gainesville, FL*
- Developing an end-to-end **AI video generation pipeline** in **ComfyUI** for the **U.S. Air Force**, producing cadet training content on sexual assault prevention.
- Scaled the pipeline on **HiPerGator HPC** using **multi-node Slurm** jobs and parallelized **CUDA** workflows, processing **500 GB** of raw video/image data to generate **200 training videos** with automated text, voice, and visual effects.
- Co-authored an oral presentation at UF's **Digital Health Symposium** (Feb. 2026): *Leveraging generative AI's ability to adapt sexual assault prevention training for Air Force Cadets: Establishing proof of concept.*

Robotics Software Engineer — Jan. 2025 – May 2026
Machine Intelligence Lab @ University of Florida — *Gainesville, FL*
- Developed a C++ **Gazebo** plugin in the **ROS2 control** framework for the SubjuGator 9 submarine's gripper, integrating JointController and JointTrajectoryController for **2-DOF** velocity and position control.
- Integrated a Water Linked **Doppler Velocity Log (DVL)** in C++, Python, and Bash on Linux, raising the localization update rate to **10 Hz** and cutting drift from **12 m/hr to 4 m/hr** in sea trials.
- Advanced SubjuGator 9 to the **RoboSub 2025 semifinals**, placing **12th of 55** international teams.

Competitions & Awards

Vobile Innovation Challenge x IGNITE: 1st Place | *Python, OpenClaw, LangChain* — Apr. 2026
- Engineered a multi-stage **prompt-chaining pipeline** that generates, evaluates, and refines creative concepts for Vobile's DreamMaker platform, enforcing consistency with **JSON-mode schemas** and few-shot exemplars.
- Developed a **sliding-window visual grounding** algorithm that generates video in **5-second segments**, re-injecting each clip's reference keyframe as a visual context anchor to eliminate cross-segment hallucination.

ITServe Synergy Startup Cube Pitch Competition: Regional Winner | *Lingua* — 2026

Technical Skills

Languages:	Python, C++, TypeScript, JavaScript, Java, SQL, Bash
AI/ML:	PyTorch, TensorFlow, LangChain, OpenAI API, CUDA, ComfyUI, OpenCV
Systems & Robotics:	Linux, Slurm (HPC), ROS2, Gazebo, NVIDIA Isaac Sim, Git
Web & Data:	Next.js, React, React Native, Node.js, Express.js, FastAPI, PostgreSQL, MongoDB, Firebase
Certifications:	AWS Certified AI Practitioner, NVIDIA DLI (Fundamentals of Deep Learning, Transformer-Based NLP)